



PROGRAMMING FUNDAMENTALS.

C++ I.

INTERVIEW III - ASCII CODE PROGRAM

Objective:

Create a C++ program that asks the user to enter one character and then displays the corresponding ASCII code for that character.

1. FILE SUBMISSION REQUIREMENT

You must submit only ONE C++ source file.

The file must be named exactly:

ASCII.cpp

Your file must compile and run successfully.

2. PROGRAM REQUIREMENTS

Your program must:

1. Ask the user to enter one character.
2. Validate that the user entered exactly ONE character.
3. Validate that the character is within the accepted ASCII range:
32 through 127
4. Display the character entered and its corresponding ASCII code.

Example:

Enter one character: A

Character entered: A

ASCII code: 65

3. INPUT VALIDATION REQUIREMENTS

The program must reject invalid input.

Invalid input includes:

1. Empty input.
2. More than one character.
3. Characters outside the accepted ASCII range.

Accepted ASCII range:

32 through 127

The program should continue asking the user for valid input until valid input is entered. When user wants to terminate the program he must input EXIT.

4. MODULARITY REQUIREMENTS

Your program must be fully modular.

This means you must separate the program into functions with clear responsibilities.

Possible functions include:

```
userInput()  
validateInput()  
getAsciiCode()  
userOutput()  
runProgram()
```

You may use different function names, but your functions must clearly show separation of concerns.

For example:

- One function should handle input.
- One function should validate input.
- One function should calculate or retrieve the ASCII code.
- One function should display the output.
- main() should be short and should mainly call another function.

5. POINTER AND STRUCT REQUIREMENTS

Pointers should be used when appropriate.

For example, you may use a pointer to pass data to a function or modify a value through a function parameter.

Structs are optional.

Use a struct only if it helps organize the program clearly. Do not use a struct just to use one.

6. DOCUMENTATION REQUIREMENTS

Your program must be fully documented.

You must include:

1. A program header comment with:

- Student name
- Course
- Assignment name
- Program description

2. Comments explaining the purpose of each function.

3. Clear comments for important logic, especially input validation.

Your documentation should help another programmer understand your program.

7. COMPILATION REQUIREMENT

Your program must compile successfully.

Programs that do not compile will receive a significant deduction.

Before submitting, test your program using different inputs.

Test examples:

A
z
5
@
space character
multiple characters such as ABC
empty input

8. EXPECTED PROGRAM BEHAVIOR

Example 1:

Enter one character: A

Character entered: A
ASCII code: 65

Example 2:

Enter one character: ABC

Invalid input. Please enter exactly one character.

Enter one character: B

Character entered: B
ASCII code: 66

Example 3:

Enter one character:

Invalid input. Please enter exactly one character.

Enter one character: C

Character entered: C
ASCII code: 67

9. IMPORTANT REMINDERS

- Submit only ASCII.cpp.
- The program must compile.
- The program must be modular.
- The program must validate input.
- The accepted ASCII range is 32 through 127.
- Use functions to separate concerns.
- Use pointers when appropriate.
- Use structs only if they improve the design.
- Fully document your program.

10. FINAL SUBMISSION

Submit your completed ASCII.cpp file in Blackboard.

Late submissions are not allowed.

Any question? Just email me.

LDV.